

Simply supported beam with uniform load

L = 5 m, q = 10 kN/m downward.

Source: examples\beams\simple-supported-beam.json

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Model overview

Nodes: 2

Members: 1

Supports: 2

Load cases: 1

Load combinations: 0

Loads: 1

Assumptions and limitations

First-order linear elastic 2D analysis. Values are reported in StructuralSolver2D internal units: displacements in m, rotations in rad, forces in kN and moments in kNm.

This PDF is a technical computation report. It is not a complete code-compliance document and does not replace engineering judgement or project-specific normative checks.

Executive summary

Metric	Entity/Member	Position	Distance	Value	Unit
Max Ux	A			0.000000	m
Max Uy	A			0.000000	m
Max Rz	A			-0.012401	rad
Max Fx	SA			0.000000	kN
Max Fy	SA			25.000000	kN
Max Mz	SA			0.000000	kNm
Max N	M1	0.000000	0.000000	-0.000000	kN
Max V	M1	0.000000	0.000000	25.000000	kN
Max M	M1	0.500000	2.500000	31.250000	kNm

Nodal displacements

Node	Ux [m]	Uy [m]	Rz [rad]
A	0.000000	0.000000	-0.012401
B	0.000000	0.000000	0.012401

Support reactions

Support	Node	Fx [kN]	Fy [kN]	Mz [kNm]
SA	A	0.000000	25.000000	0.000000
SB	B	0.000000	25.000000	0.000000

Member end forces

Member	N1 [kN]	V1 [kN]	M1 [kNm]	N2 [kN]	V2 [kN]	M2 [kNm]
M1	0.000000	25.000000	0.000000	0.000000	25.000000	0.000000

Internal-force samples

Member	Pos	x [m]	N [kN]	V [kN]	M [kNm]
M1	0.000000	0.000000	-0.000000	25.000000	0.000000
M1	0.050000	0.250000	-0.000000	22.500000	5.937500
M1	0.100000	0.500000	-0.000000	20.000000	11.250000
M1	0.150000	0.750000	-0.000000	17.500000	15.937500
M1	0.200000	1.000000	-0.000000	15.000000	20.000000
M1	0.250000	1.250000	-0.000000	12.500000	23.437500
M1	0.300000	1.500000	-0.000000	10.000000	26.250000

M1	0.350000	1.750000	-0.000000	7.500000	28.437500
M1	0.400000	2.000000	-0.000000	5.000000	30.000000
M1	0.450000	2.250000	-0.000000	2.500000	30.937500
M1	0.500000	2.500000	-0.000000	0.000000	31.250000
M1	0.550000	2.750000	-0.000000	-2.500000	30.937500
M1	0.600000	3.000000	-0.000000	-5.000000	30.000000
M1	0.650000	3.250000	-0.000000	-7.500000	28.437500
M1	0.700000	3.500000	-0.000000	-10.000000	26.250000
M1	0.750000	3.750000	-0.000000	-12.500000	23.437500
M1	0.800000	4.000000	-0.000000	-15.000000	20.000000
M1	0.850000	4.250000	-0.000000	-17.500000	15.937500
M1	0.900000	4.500000	-0.000000	-20.000000	11.250000
M1	0.950000	4.750000	-0.000000	-22.500000	5.937500
M1	1.000000	5.000000	-0.000000	-25.000000	0.000000

Displacement samples

Member	Pos	x [m]	GUx [m]	GUy [m]
M1	0.000000	0.000000	0.000000	0.000000
M1	0.050000	0.250000	0.000000	-0.002945
M1	0.100000	0.500000	0.000000	-0.005580
M1	0.150000	0.750000	0.000000	-0.007906
M1	0.200000	1.000000	0.000000	-0.009921
M1	0.250000	1.250000	0.000000	-0.011626
M1	0.300000	1.500000	0.000000	-0.013021
M1	0.350000	1.750000	0.000000	-0.014106
M1	0.400000	2.000000	0.000000	-0.014881
M1	0.450000	2.250000	0.000000	-0.015346
M1	0.500000	2.500000	0.000000	-0.015501
M1	0.550000	2.750000	0.000000	-0.015346
M1	0.600000	3.000000	0.000000	-0.014881
M1	0.650000	3.250000	0.000000	-0.014106
M1	0.700000	3.500000	0.000000	-0.013021
M1	0.750000	3.750000	0.000000	-0.011626
M1	0.800000	4.000000	0.000000	-0.009921
M1	0.850000	4.250000	0.000000	-0.007906
M1	0.900000	4.500000	0.000000	-0.005580
M1	0.950000	4.750000	0.000000	-0.002945
M1	1.000000	5.000000	0.000000	0.000000